Missed ACS Codes Review

Nine knowledge codes, one lesson

Cessna 172S NAV III (G1000)

FAA-S-ACS-25

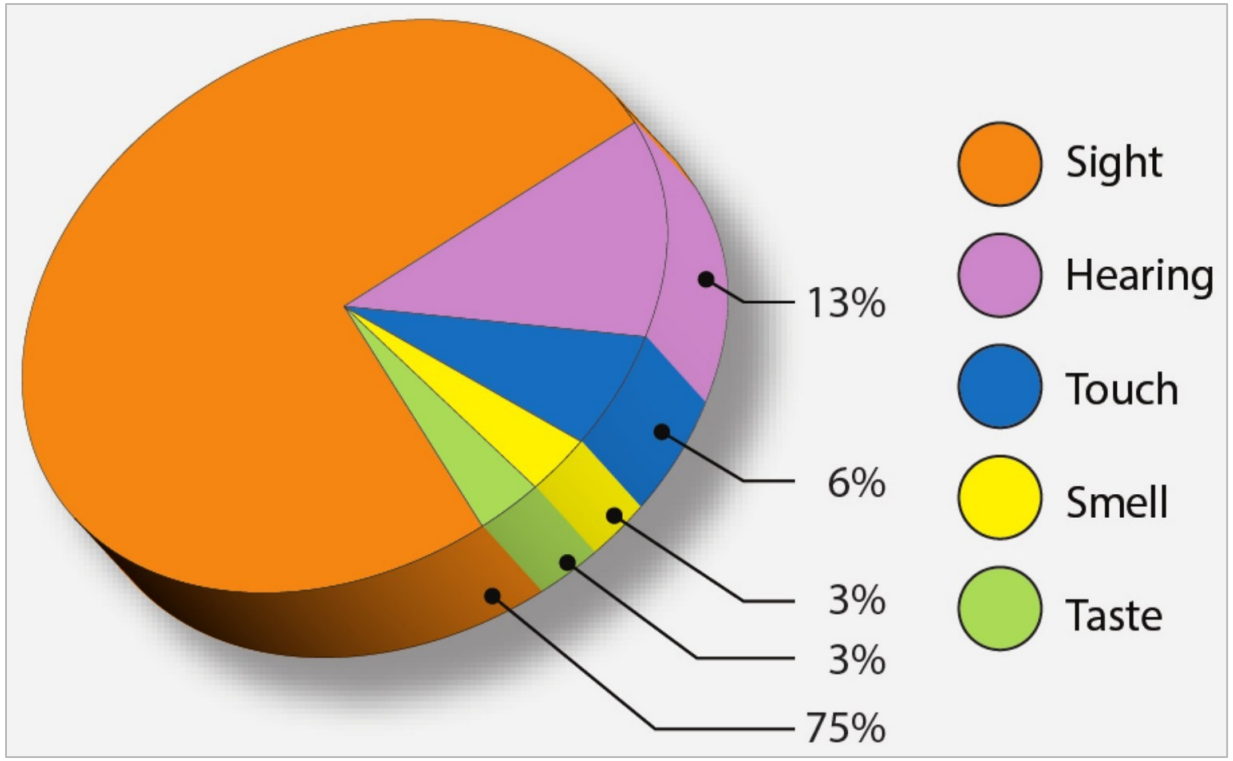
[Name], CFI Applicant



PHAK Fig 16-29

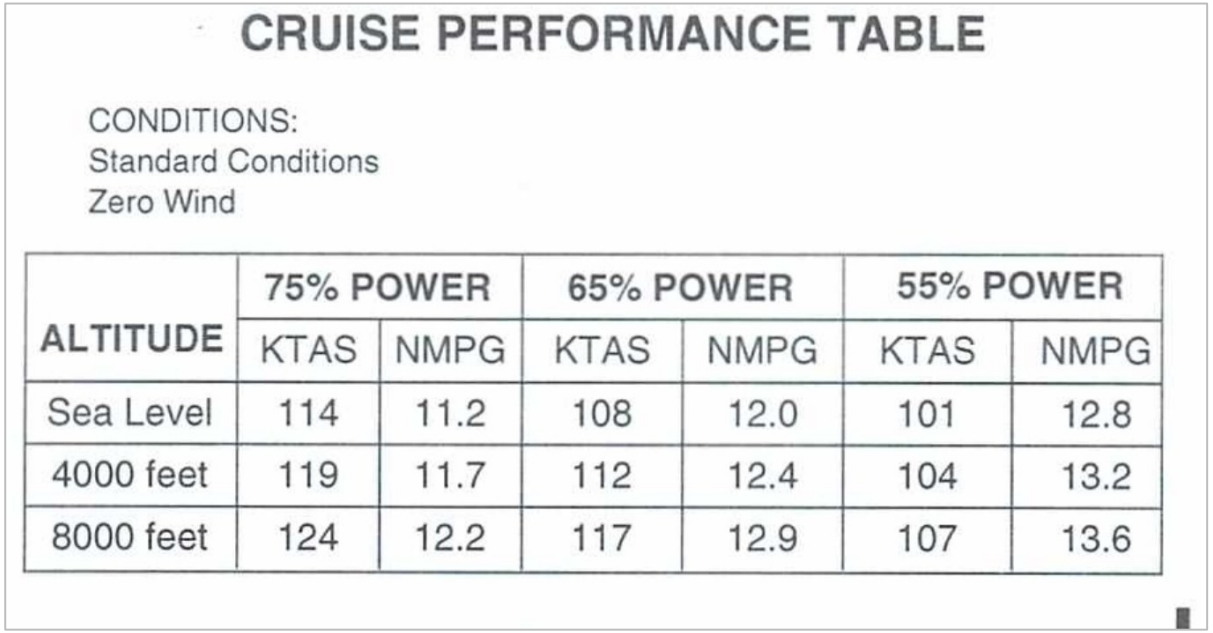
Objective

- Explain each missed code
- Use FAA wording
- Tie it to the C172S
- Teach it back clearly



The Nine Codes

Code	Topic
FI.I.B.K3	Perceptions and insight
AI.II.D.K2	Stability and control
AI.II.E.K1c	Powerplant and propeller
AI.II.F.K3	Weight and balance terms
AI.II.H.K1	Ground-based navigation
AI.II.I.K3	Plotting a course
AI.II.I.K4	Power setting selection
AI.III.B.K1b	Inspections and logbooks
AI.XII.D.K1	ELT



CRUISE PERFORMANCE TABLE

CONDITIONS:
Standard Conditions
Zero Wind

ALTITUDE	75% POWER		65% POWER		55% POWER	
	KTAS	NMPG	KTAS	NMPG	KTAS	NMPG
Sea Level	114	11.2	108	12.0	101	12.8
4000 feet	119	11.7	112	12.4	104	13.2
8000 feet	124	12.2	117	12.9	107	13.6

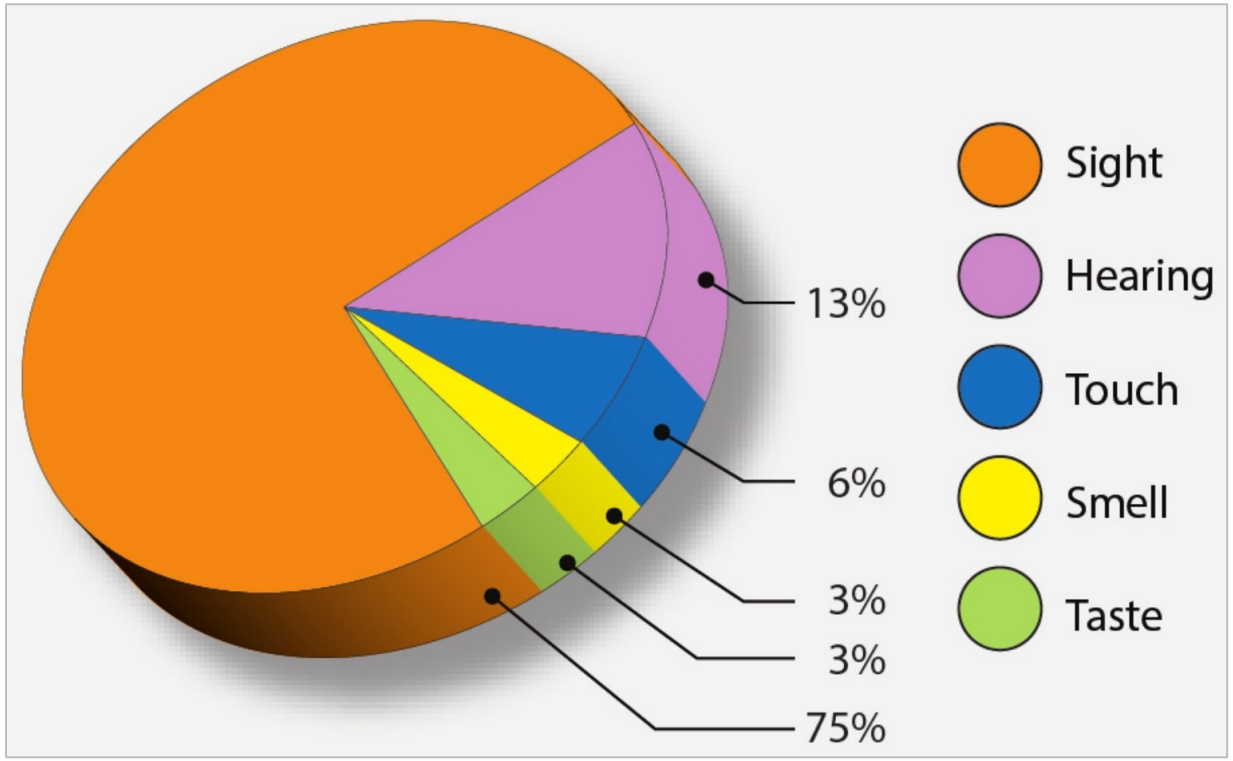
Definition: Perceptions

Perceptions

“Initially, all learning comes from perceptions, which are directed to the brain by one or more of the five senses: sight, hearing, touch, smell, and taste.”

More than one sense

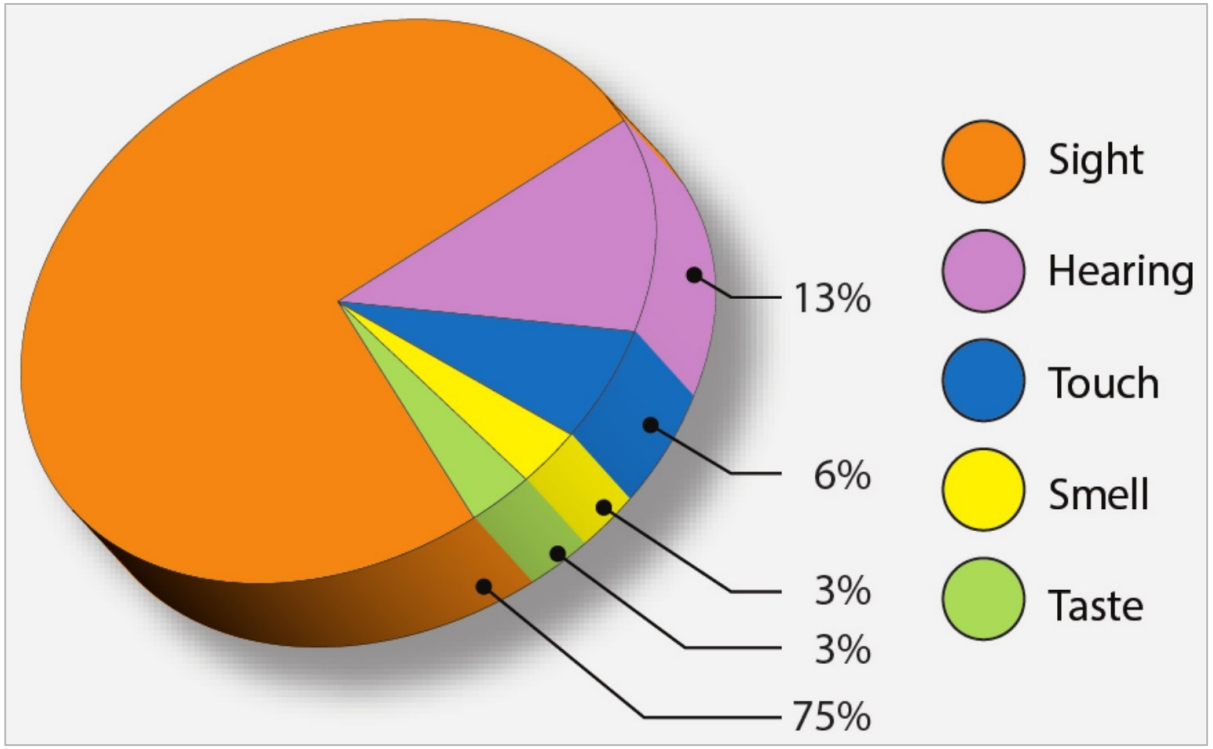
“learning occurs most rapidly when information is received through more than one sense.”



AIH 3-7, AIH Fig 3-5

Factors That Affect Perception

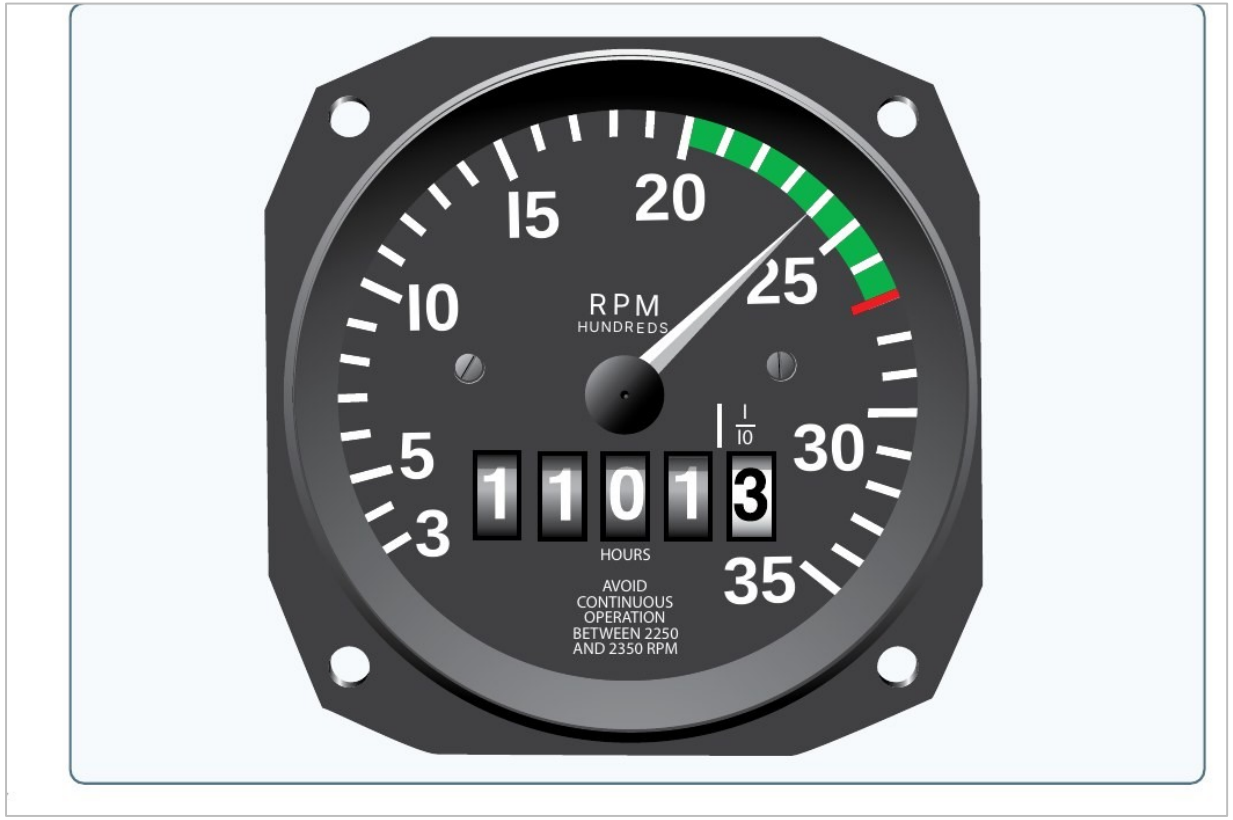
Factor	Plain words
Physical organism	Senses must work
Goals and values	What they care about
Self-concept	Confident or not
Time and opportunity	Time and right order
Element of threat	Fear narrows focus



Definition: Insight

Insight

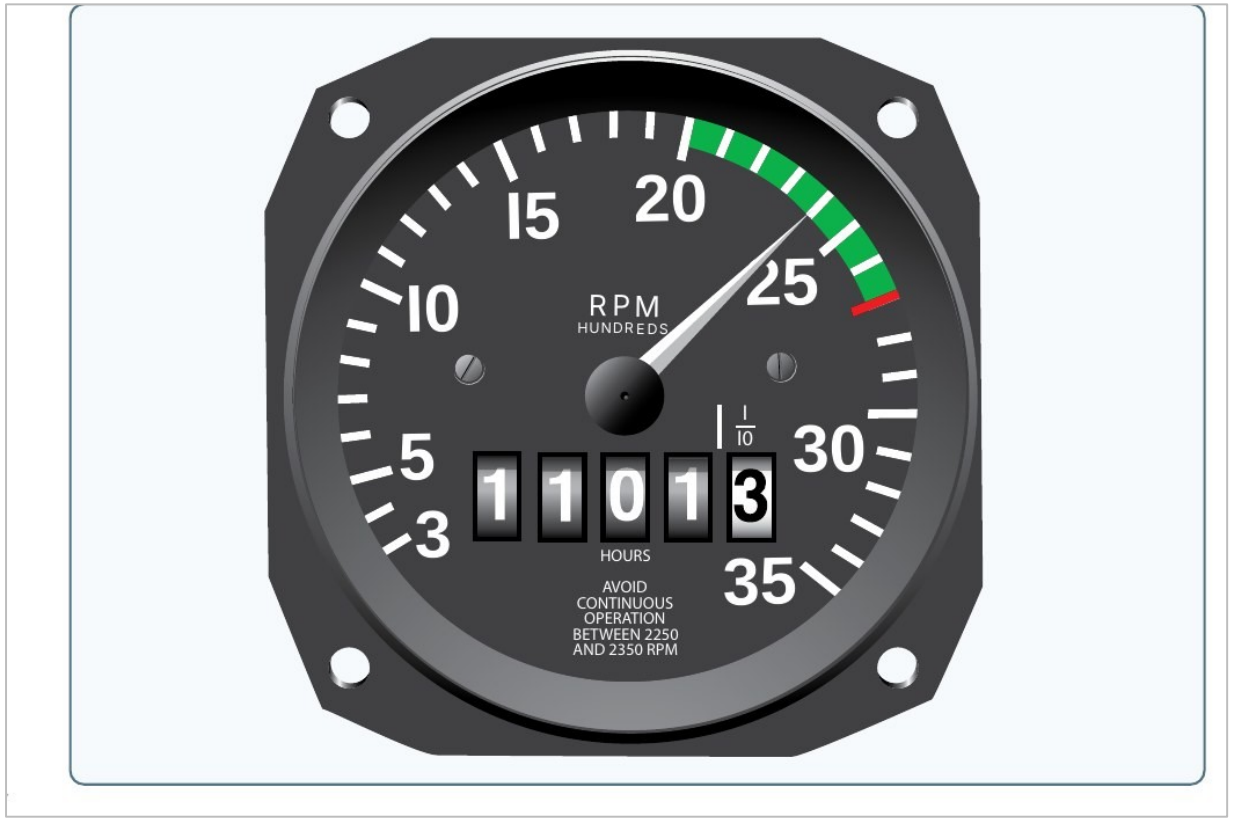
“Insight involves the grouping of perceptions into meaningful wholes. Creating insight is one of the instructor's major responsibilities.”



AIH 3-9, PHAK Fig 7-8

How Instructors Build Insight

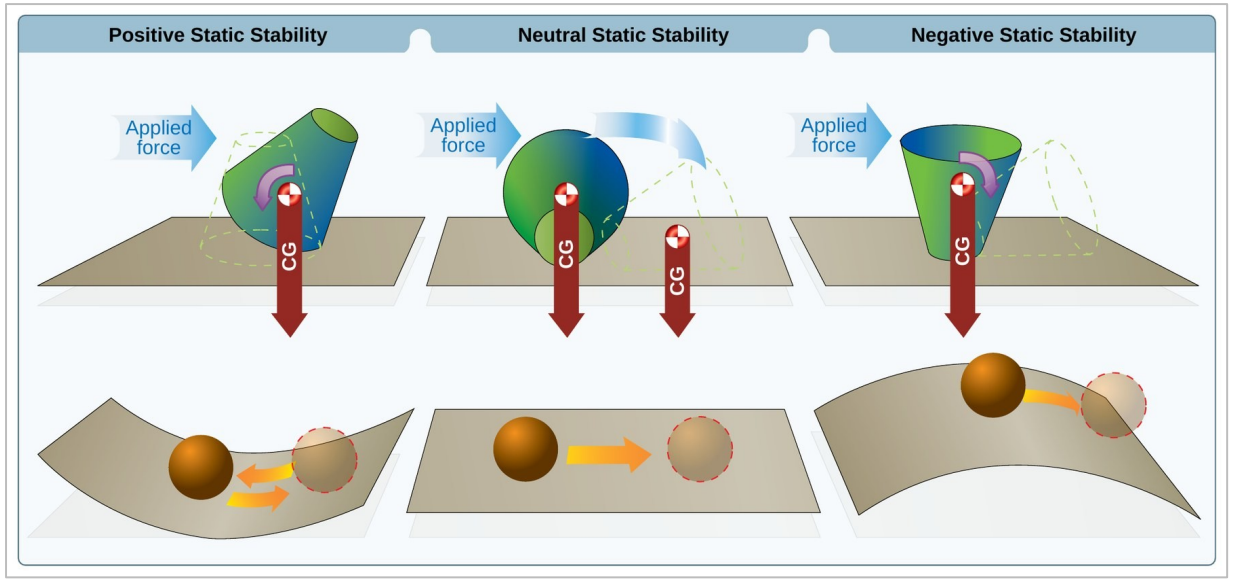
- Point out relationships
- Keep the learner receptive
- Nonthreatening environment
- Protect a good self-concept



Definition: Stability

Stability

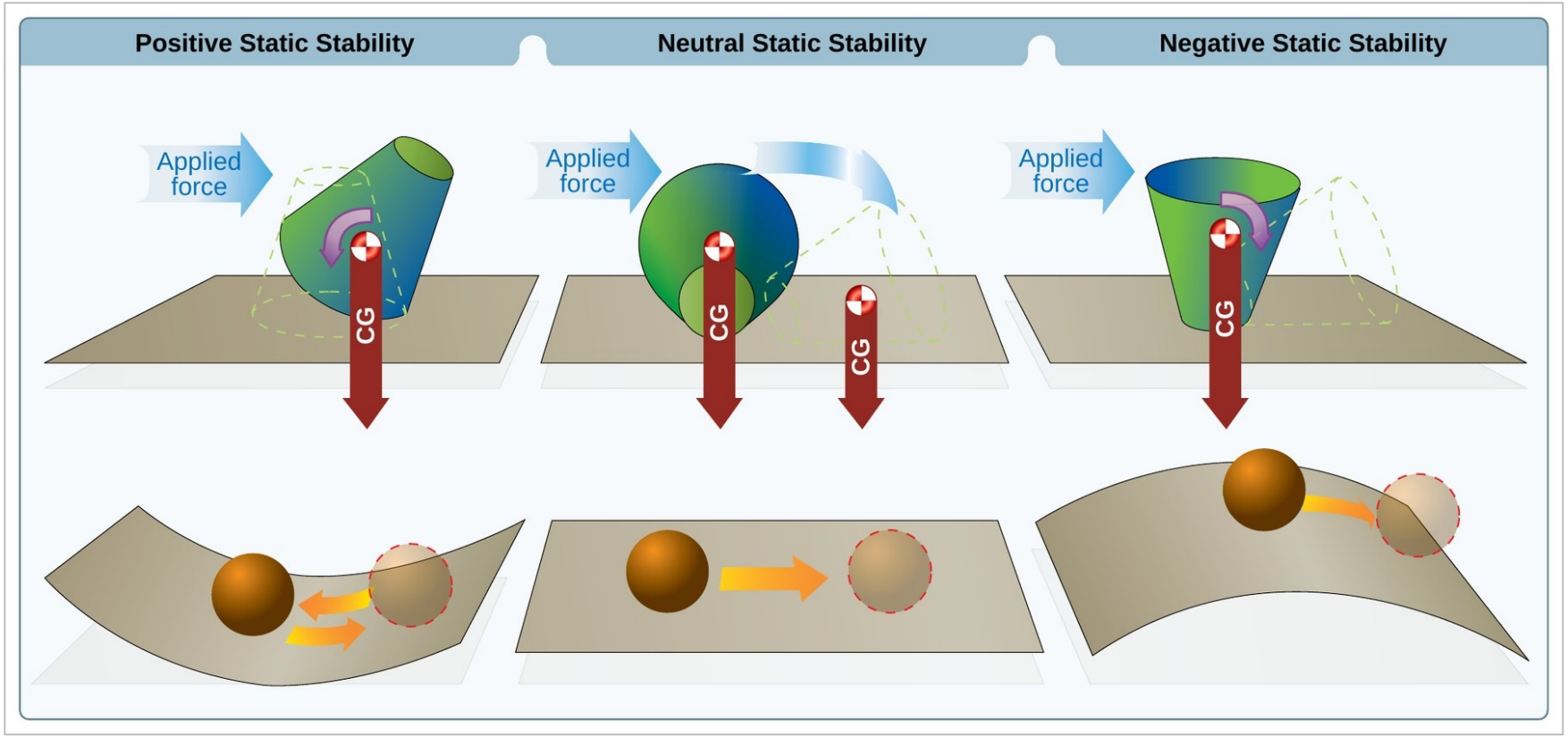
“Stability is the inherent quality of an aircraft to correct for conditions that may disturb its equilibrium and to return to or to continue on the original flight path. It is primarily an aircraft design characteristic.”



PHAK 5-14, PHAK Fig 5-21

Static Stability

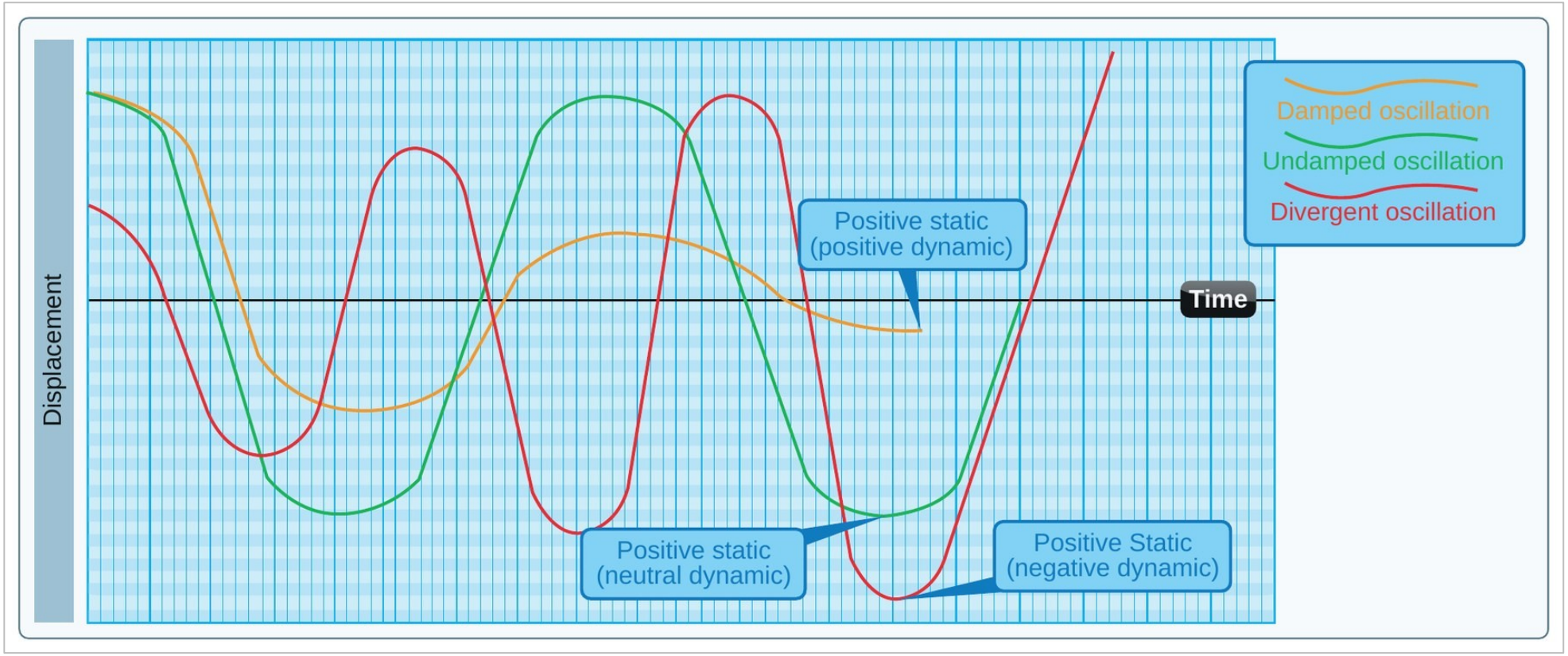
- Initial tendency after disturbance
- Positive: returns to equilibrium
- Neutral: stays in new condition
- Negative: continues away



PHAK 5-14, PHAK Fig 5-21

Dynamic Stability

- Response over time
- Positive: motion gets smaller
- Neutral: motion stays the same
- Negative: motion grows



PHAK 5-14, PHAK 5-15, PHAK Fig 5-22

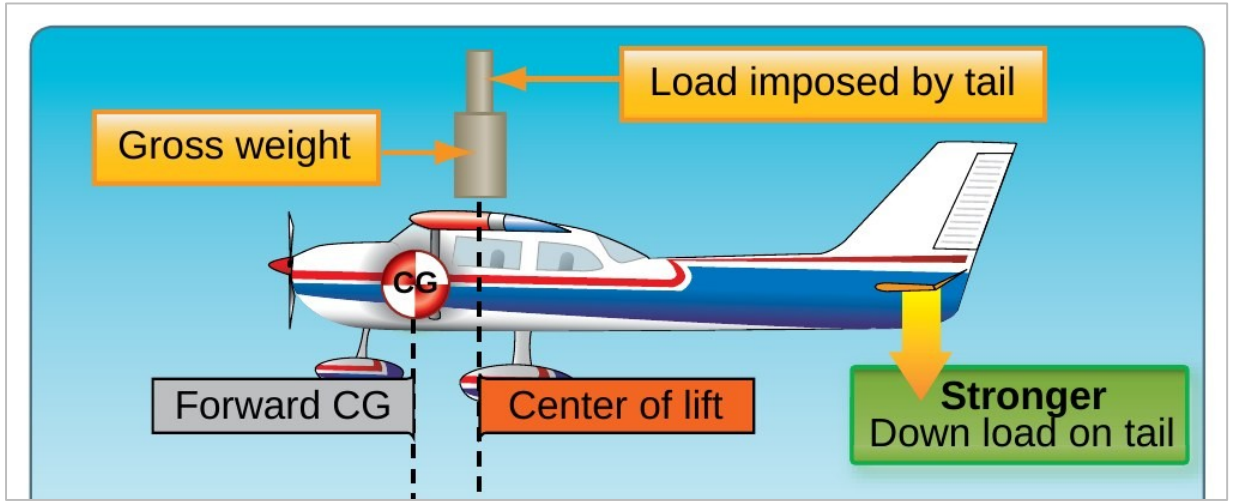
Definitions: Maneuverability and Controllability

Maneuverability

“the quality of an aircraft that permits it to be maneuvered easily and to withstand the stresses imposed by maneuvers. It is governed by the aircraft's weight, inertia, size and location of flight controls, structural strength, and powerplant.”

Controllability

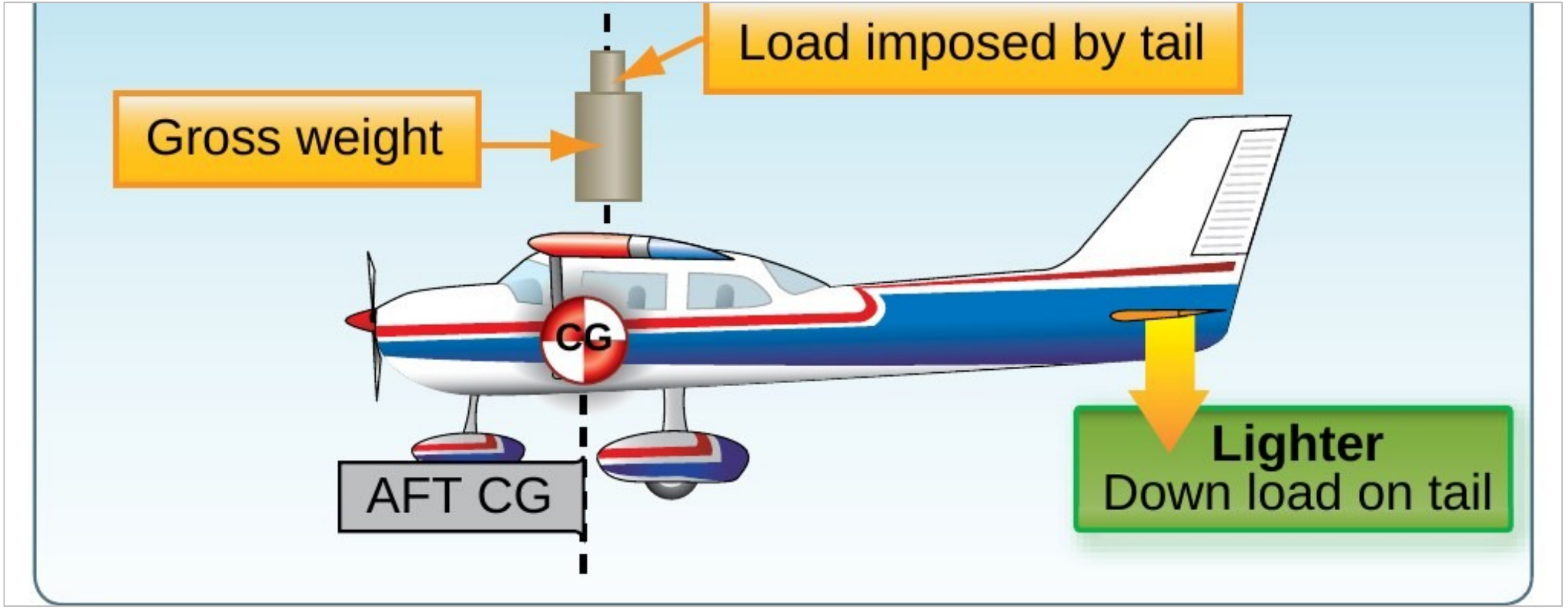
“the capability of an aircraft to respond to the pilot's control, especially with regard to flight path and attitude.”



PHAK 5-15, PHAK Fig 5-63

Putting the Three Together

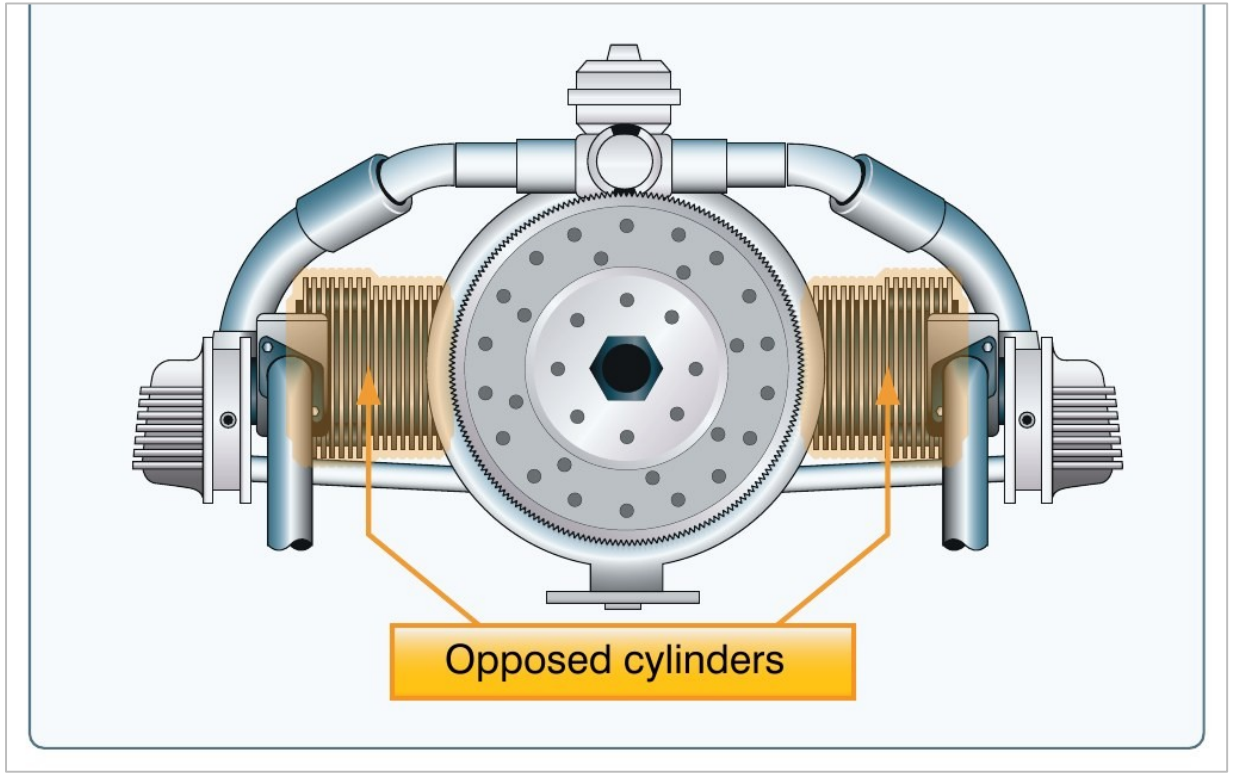
- Stability: wants to return
- Maneuverability: maneuvers easily
- Controllability: answers the pilot
- CG location changes all three



PHAK 5-15, PHAK 5-43, PHAK 5-44

C172S Engine

- Lycoming IO-360-L2A
- 180 hp at 2,700 RPM
- Four cylinder, horizontally opposed
- Fuel injected, air cooled



Engine Controls

- Throttle: fuel-air flow, RPM
- Mixture: red knob
- Mixture full aft: idle cutoff
- No propeller control

ENGINE

The airplane is powered by a horizontally opposed, four cylinder, overhead valve, air-cooled, fuel-injected engine with a wet sump lubrication system. The engine is a Lycoming Model IO-360-L2A and is rated at 180 horsepower at 2700 RPM. Major accessories include a starter and belt driven alternator mounted on the front of the engine, and dual magnetos, dual vacuum pumps, and a full flow oil filter mounted on the rear of the engine accessory case.

POH 7-28, PHAK 7-5

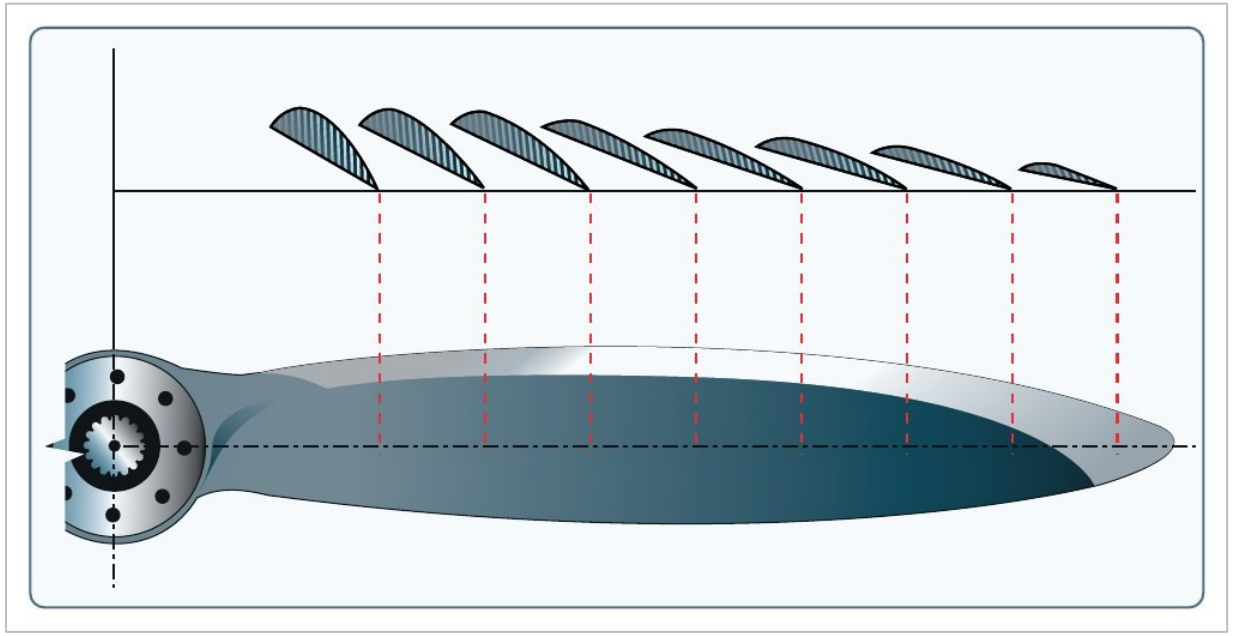
Definitions: Propeller and Fixed Pitch

Propeller

“The propeller is a rotating airfoil, subject to induced drag, stalls, and other aerodynamic principles that apply to any airfoil.”

Fixed-pitch propeller

“A propeller with fixed blade angles is a fixed-pitch propeller. The pitch of this propeller is set by the manufacturer and cannot be changed.”



C172S Propeller

- Two blade, fixed pitch
- 76 inch forged aluminum
- Best at one speed, rpm
- Tachometer shows power

PROPELLER

The airplane is equipped with a two bladed, fixed pitch, one-piece forged aluminum alloy propeller which is anodized to retard corrosion. The propeller is 76 inches in diameter.

POH 7-36, PHAK 7-5, PHAK 5-29

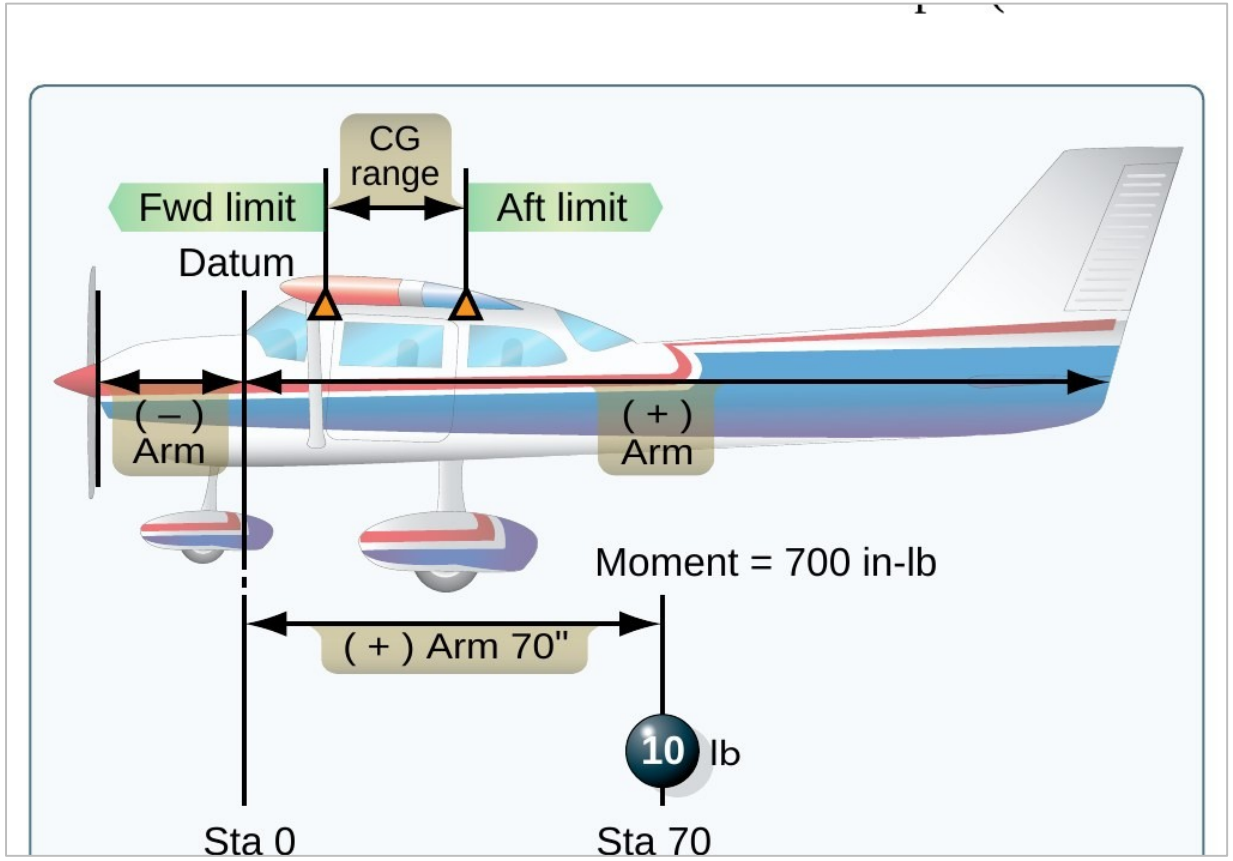
Definitions: Datum and Arm

Datum (reference datum)

“an imaginary vertical plane or line from which all measurements of arm are taken. The datum is established by the manufacturer. Once the datum has been selected, all moment arms and the location of CG range are measured from this point.”

Arm (moment arm)

“the horizontal distance in inches from the reference datum line to the CG of an item. The algebraic sign is plus (+) if measured aft of the datum and minus (−) if measured forward of the datum.”



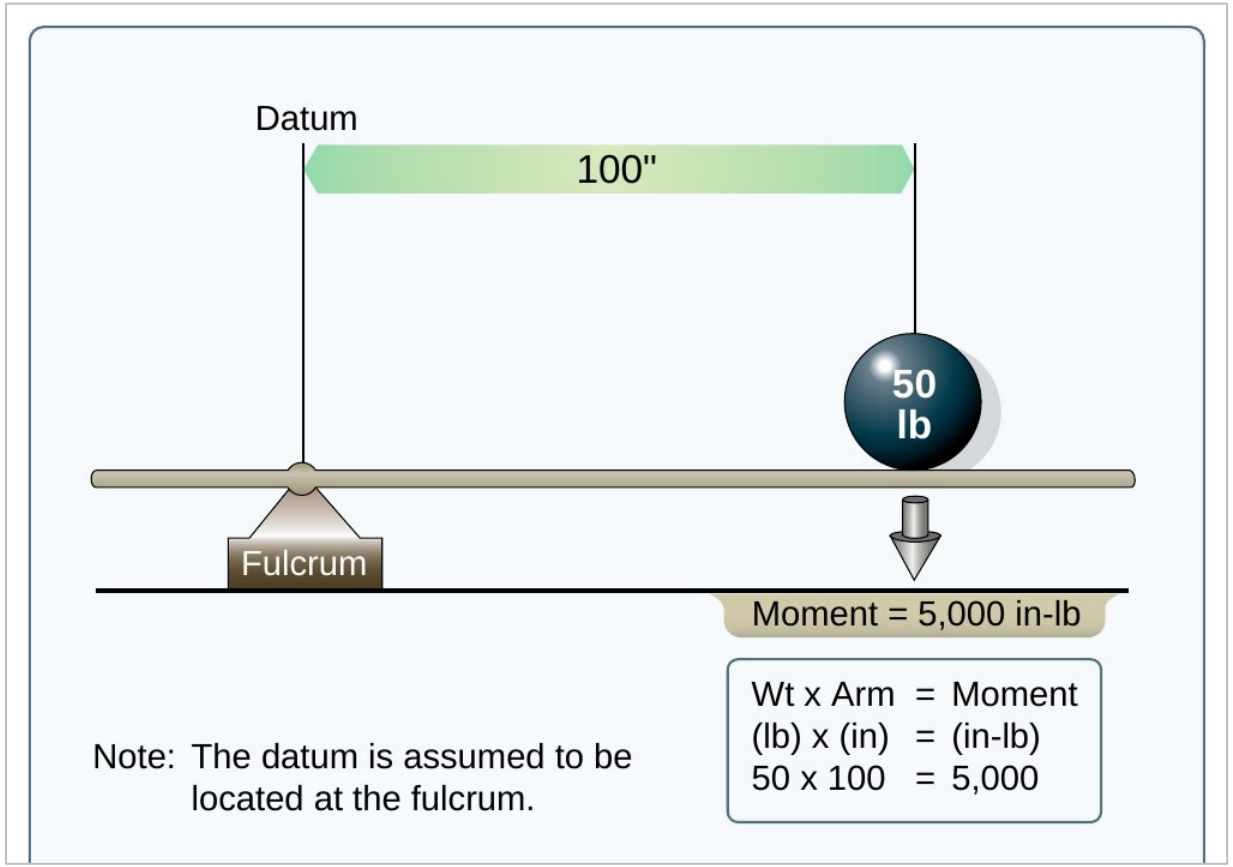
Definitions: Moment and CG

Moment

“the product of the weight of an item multiplied by its arm. Moments are expressed in pound-inches (in-lb).”

Center of gravity (CG)

“the point about which an aircraft would balance if it were possible to suspend it at that point.”



Definitions: Basic Empty Weight and Useful Load

Basic empty weight (GAMA)

“the standard empty weight plus the weight of optional and special equipment that have been installed.”

Useful load

“the weight of the pilot, copilot, passengers, baggage, usable fuel, and drainable oil. It is the basic empty weight subtracted from the maximum allowable gross weight.”

SAMPLE LOADING PROBLEM

ITEM DESCRIPTION	WEIGHT AND MOMENT TABULATION			
	SAMPLE AIRPLANE		YOUR AIRPLANE	
	Weight (Lbs.)	Moment (Lb-ins. /1000)	Weight (Lbs.)	Moment (Lb-ins. /1000)
1. Basic Empty Weight (Use the data pertaining to your airplane as it is presently equipped. Includes unusable fuel and full oil)	1642	62.6		
2. Usable Fuel (At 6 Lbs./Gal.)				
Standard Fuel 53 Gallons Maximum				
Reduced Fuel (30 Gallons)	180	8.6		
3. Pilot and Front Passenger (FS 32 to 50)	340	12.6		
4. Rear Passengers (FS 74)	340	24.8		
5.* Baggage "A" (FS 82 to 108) 120 Pounds Maximum	56	4.6		
6.* Baggage "B" (FS 108 to 142) 50 Pounds Maximum				
7. **RAMP WEIGHT AND MOMENT**	2558	113.2		
8. Fuel allowance for engine start, taxi and runup	-8.0	-0.4		
9. **TAKEOFF WEIGHT AND MOMENT** (Subtract Step 8 from Step 7)	2550	112.8		

10. Locate this point (2550 at 112.8) on the Center-of-Gravity Moment Envelope, and since this point falls within the envelope, the loading is acceptable.

* The maximum allowable combined weight capacity for baggage in areas A and B is 120 pounds.

PHAK 10-4, PHAK 10-5, POH Fig 6-3

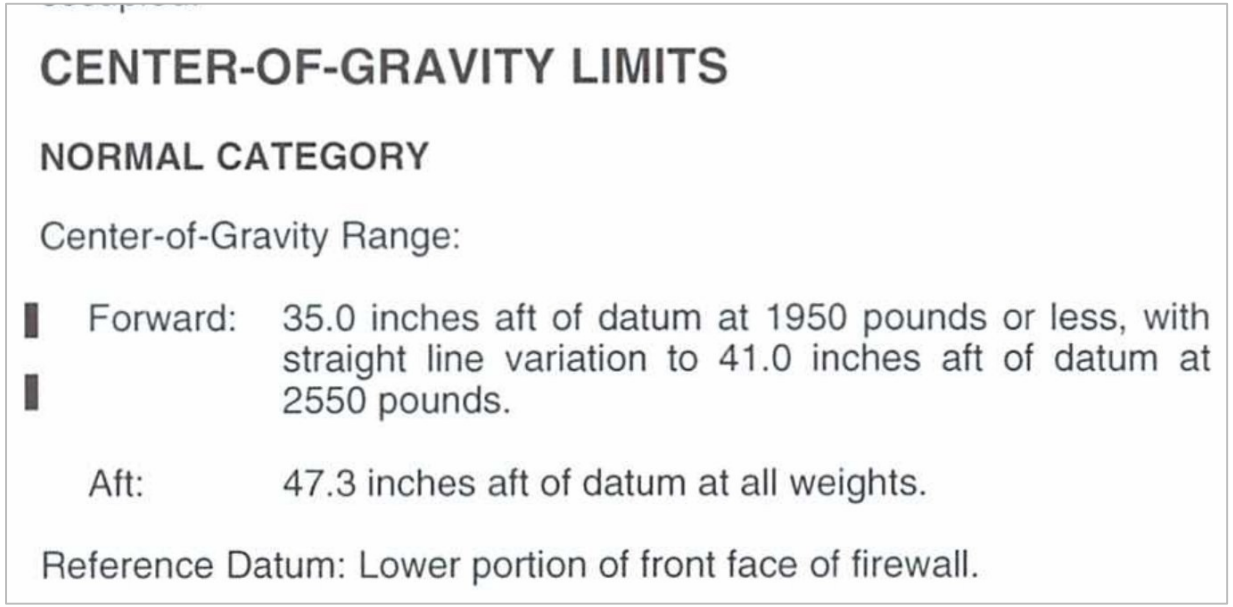
Definitions: CG Limits and Maximum Weight

CG limits

“the specified forward and aft points within which the CG must be located during flight. These limits are indicated on pertinent aircraft specifications.”

Maximum weight

“the maximum authorized weight of the aircraft and all of its equipment as specified in the TCDS for the aircraft.”



CENTER-OF-GRAVITY LIMITS

NORMAL CATEGORY

Center-of-Gravity Range:

Forward:	35.0 inches aft of datum at 1950 pounds or less, with straight line variation to 41.0 inches aft of datum at 2550 pounds.
Aft:	47.3 inches aft of datum at all weights.

Reference Datum: Lower portion of front face of firewall.

C172S Weight and Balance Numbers

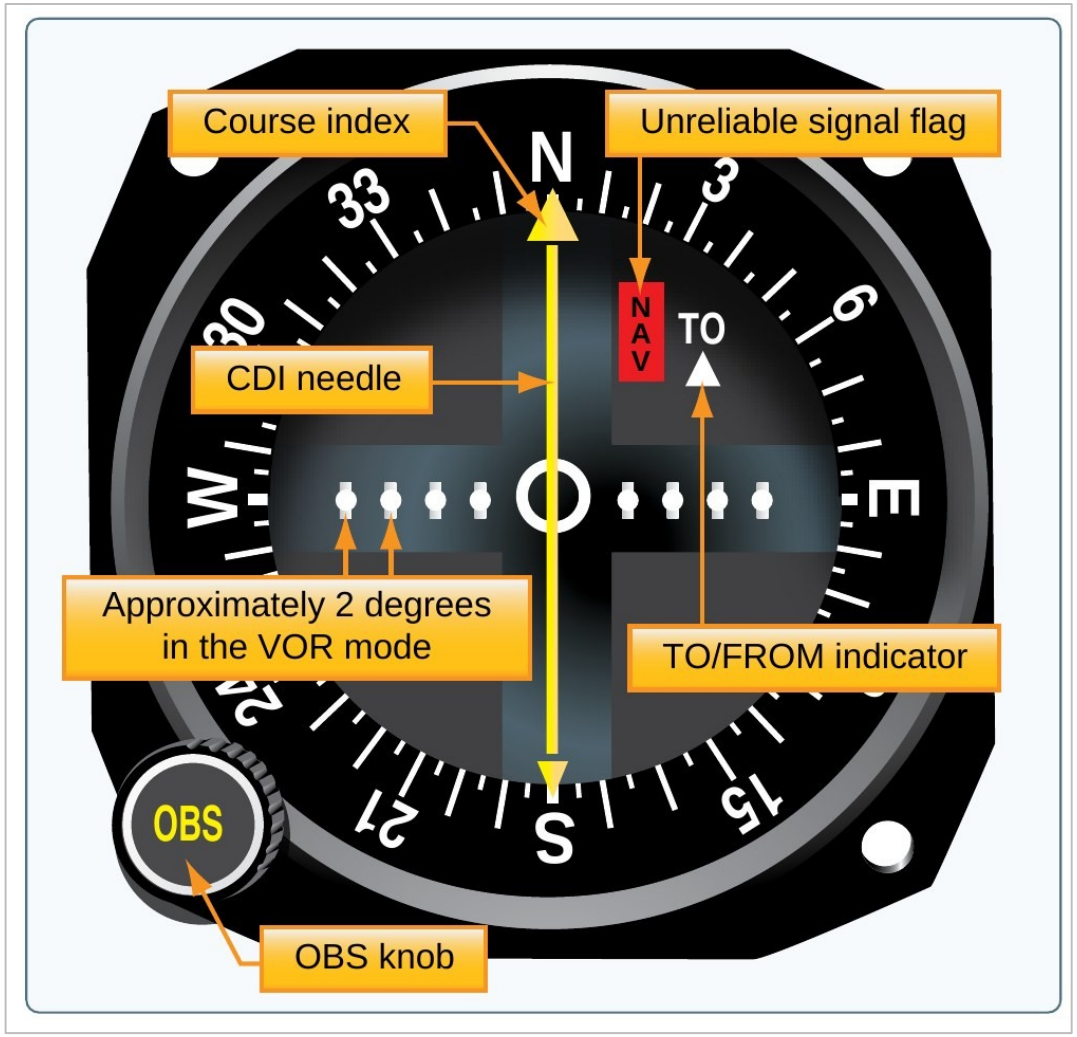
- Datum: front face of firewall
- Max takeoff: 2,550 lb
- CG: 35.0 to 47.3 in
- CG = moment / weight

C.G. Range	Normal Category	
	(1) Aft Limits	47.3 inches aft of datum at 2,550 pounds or less.
	(2) Forward Limits	Linear variation from 41.0 inches aft of datum at 2,550 pounds to 35.0 inches aft of datum at 1,950 pounds. 35.0 inches aft of datum at 1,950 pounds or less.
	Utility Category	
	(1) Aft Limits	40.5 inches aft of datum at 2,200 pounds or less.
	(2) Forward Limits	Linear variation from 37.5 inches aft of datum at 2,200 pounds to 35.0 inches aft of datum at 1,950 pounds. 35.0 inches aft of datum at 1,950 pounds or less.
Empty Wt. C.G. Range	None	
Reference Datum	Lower portion of front face of firewall	
MAC	58.8 inches; Leading edge of MAC 25.9 inches aft of datum	
Leveling Means	Left side of Tailcone at 108.0 inches and 142.0 inches aft of datum	

TCDS 3A12 p30, POH 2-8, PHAK 10-7

VOR: Identification and Accuracy

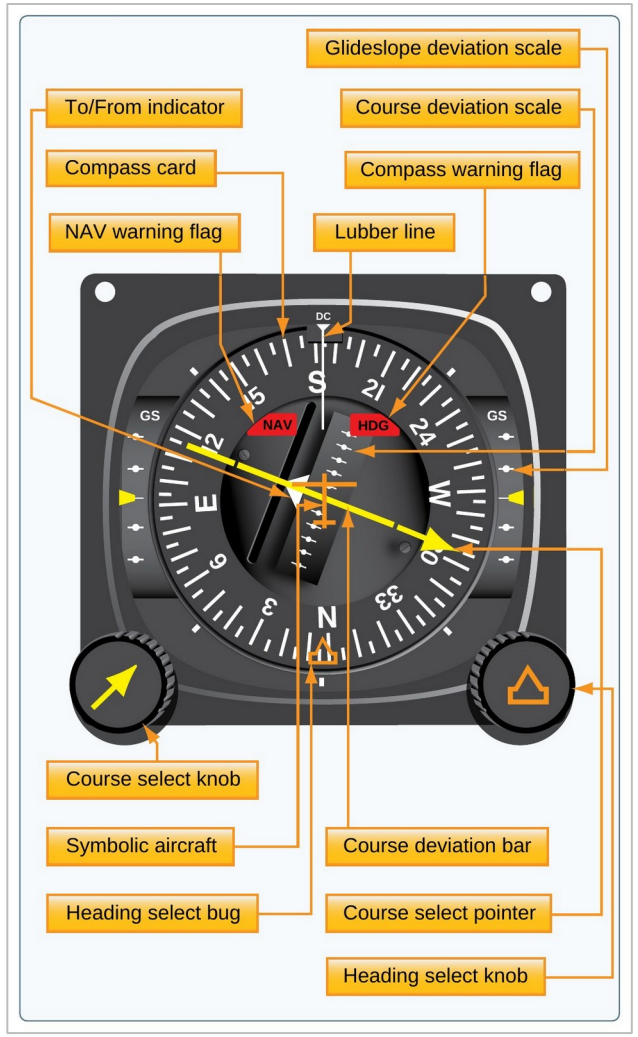
- 108.0 to 117.95 MHz
- Line of sight only
- Identify by Morse code
- Accuracy: about 1 degree



AIM 1-1-3, PHAK Fig 16-29

Orientation: CDI and HSI

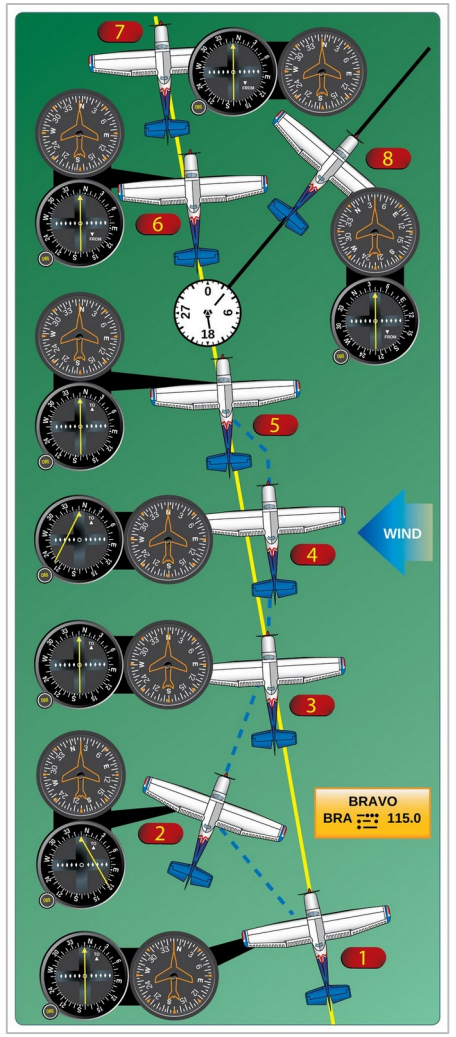
- OBS sets the course
- Needle shows course deviation
- TO/FROM shows station side
- G1000 shows an HSI



PHAK 16-23, PHAK 16-24, PHAK 16-26, PHAK Fig 16-30

Course Determination: Tracking

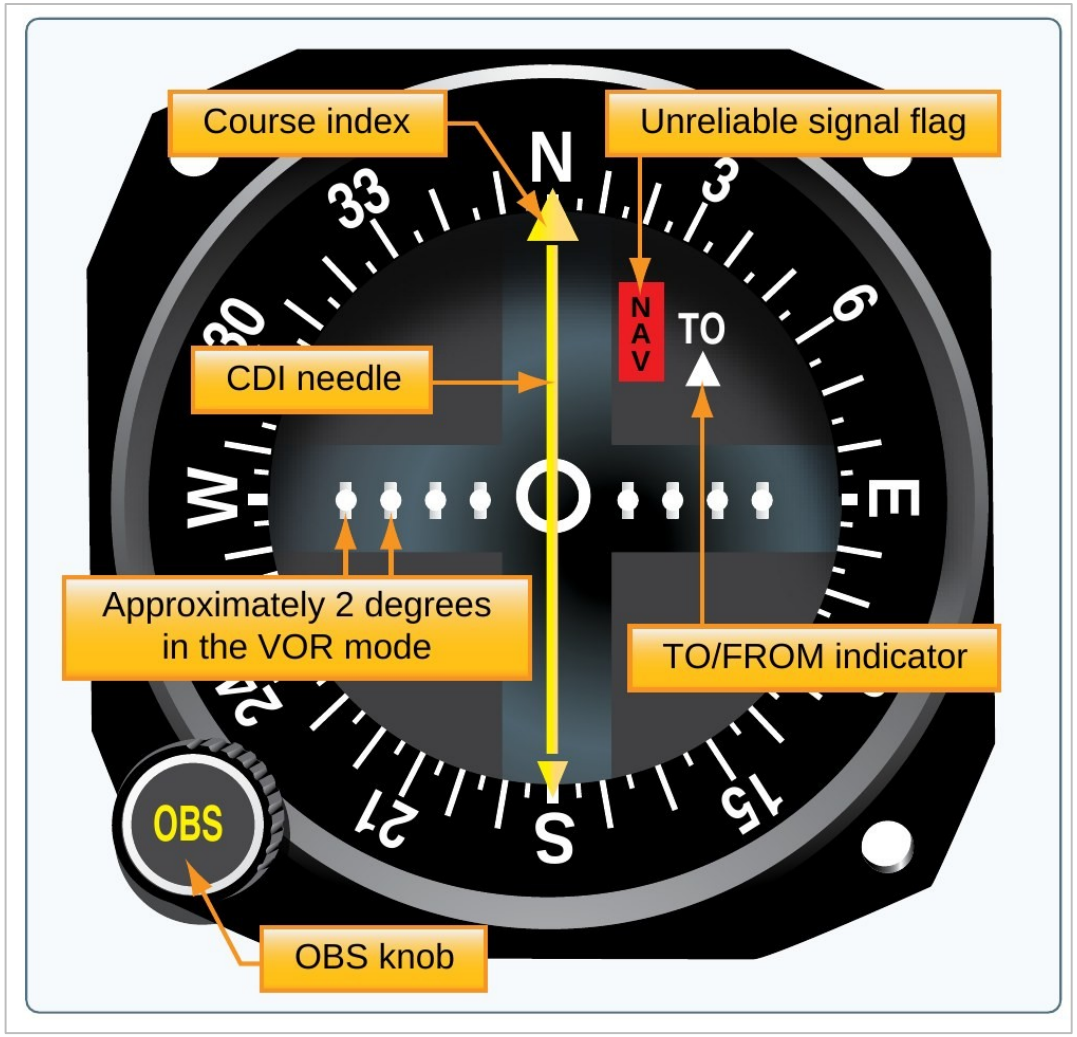
- Needle shows the course
- Turn toward needle, intercept
- Hold a wind correction
- Wait on small wiggles



PHAK 16-25, PHAK 16-26, PHAK Fig 16-32

Tests and Regulations

- VOT: 0 FROM or 180 TO
- Ground check: 4 degrees
- Airborne check: 6 degrees
- IFR: checked within 30 days



NDB and ADF

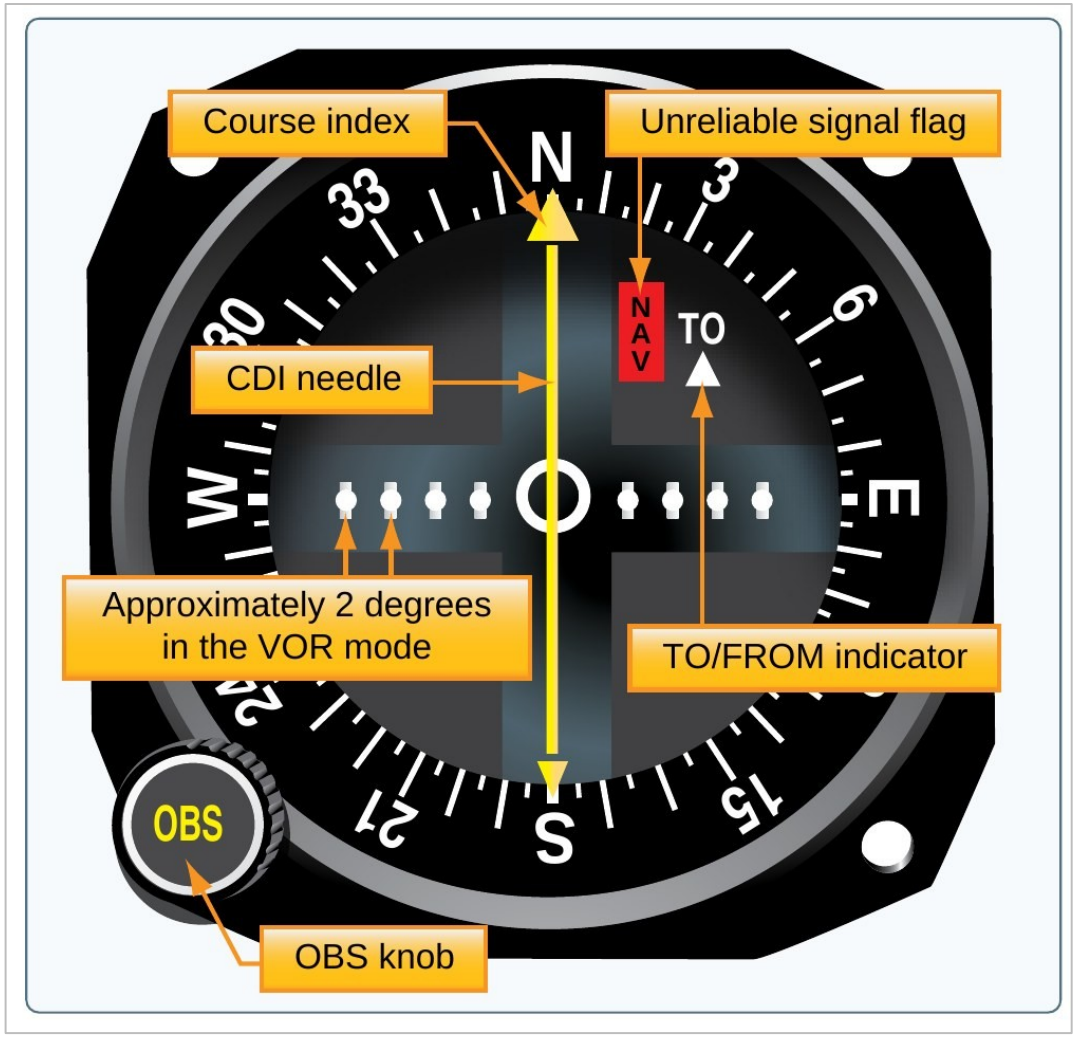
- 190 to 535 kHz
- ADF needle points to station
- No warning flag
- Keep listening to the ID



AIM 1-1-2, PHAK 16-25, PHAK Fig 16-31

Interference and Signal Integrity

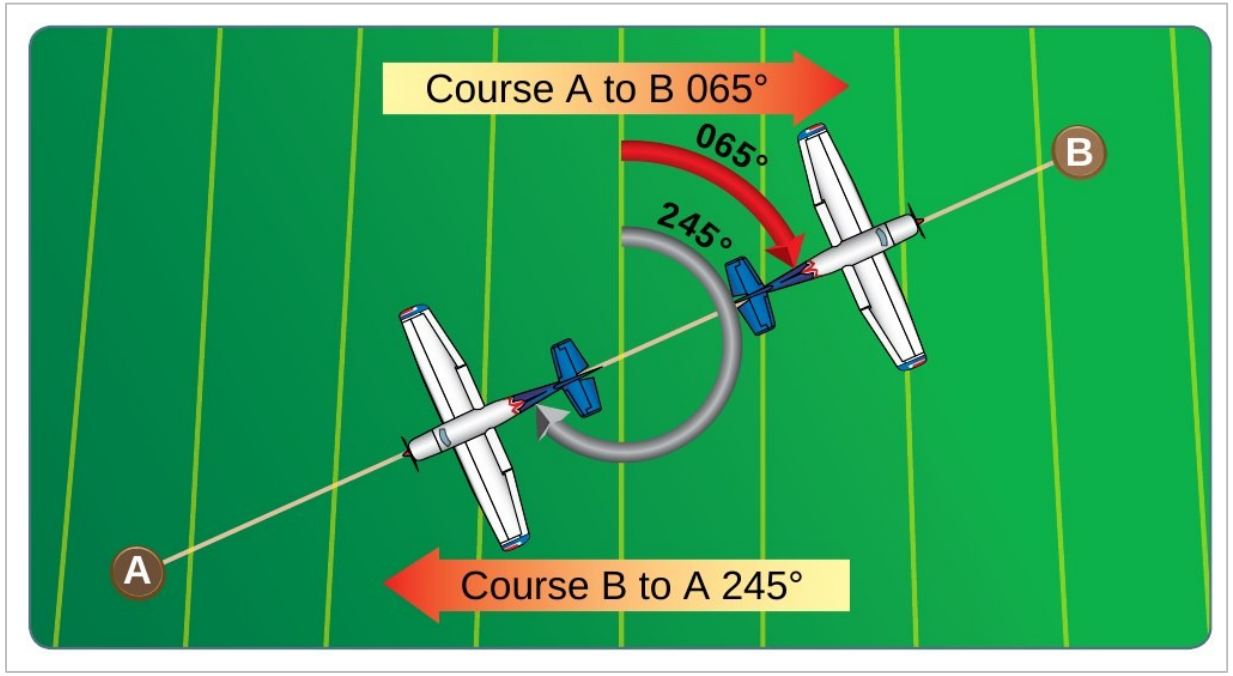
- Prop RPM can swing CDI
- Change RPM slightly
- Flag means unreliable signal
- NDB: lightning, night static



Definition: True Course

True course (TC)

“The course measured on the chart is known as the true course (TC). This is the direction measured by reference to a meridian or true north (TN).”



PHAK 16-5, PHAK Fig 16-7

Plotting the Course

- Line: airport center to center
- Measure at midpoint meridian
- Plotter gives true course
- Measure distance in NM

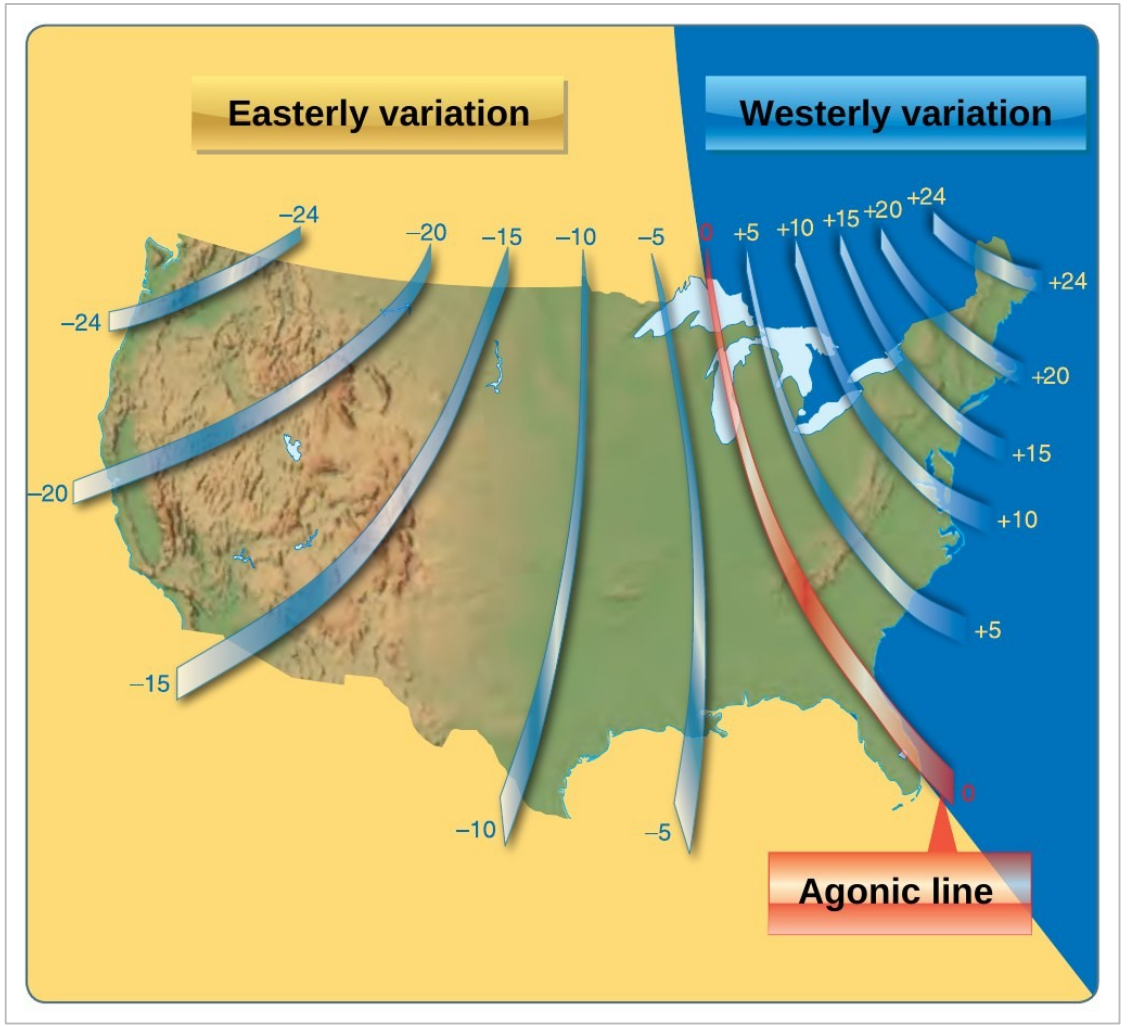


PHAK 16-5, PHAK 16-12, PHAK 16-18, PHAK Fig 16-25

Definition: Variation

Variation

“Variation is the angle between TN and magnetic north (MN). It is expressed as east variation or west variation depending upon whether MN is to the east or west of TN.”



PHAK 16-6, PHAK Fig 16-9

True Course to Magnetic Course

- Use nearest isogonic line
- West variation: add
- East variation: subtract
- East is least, west is best



Choosing a Power Setting

- Start with POH cruise table
- Pick altitude with best winds
- Low power saves fuel
- High power: faster, more fuel

CRUISE PERFORMANCE TABLE

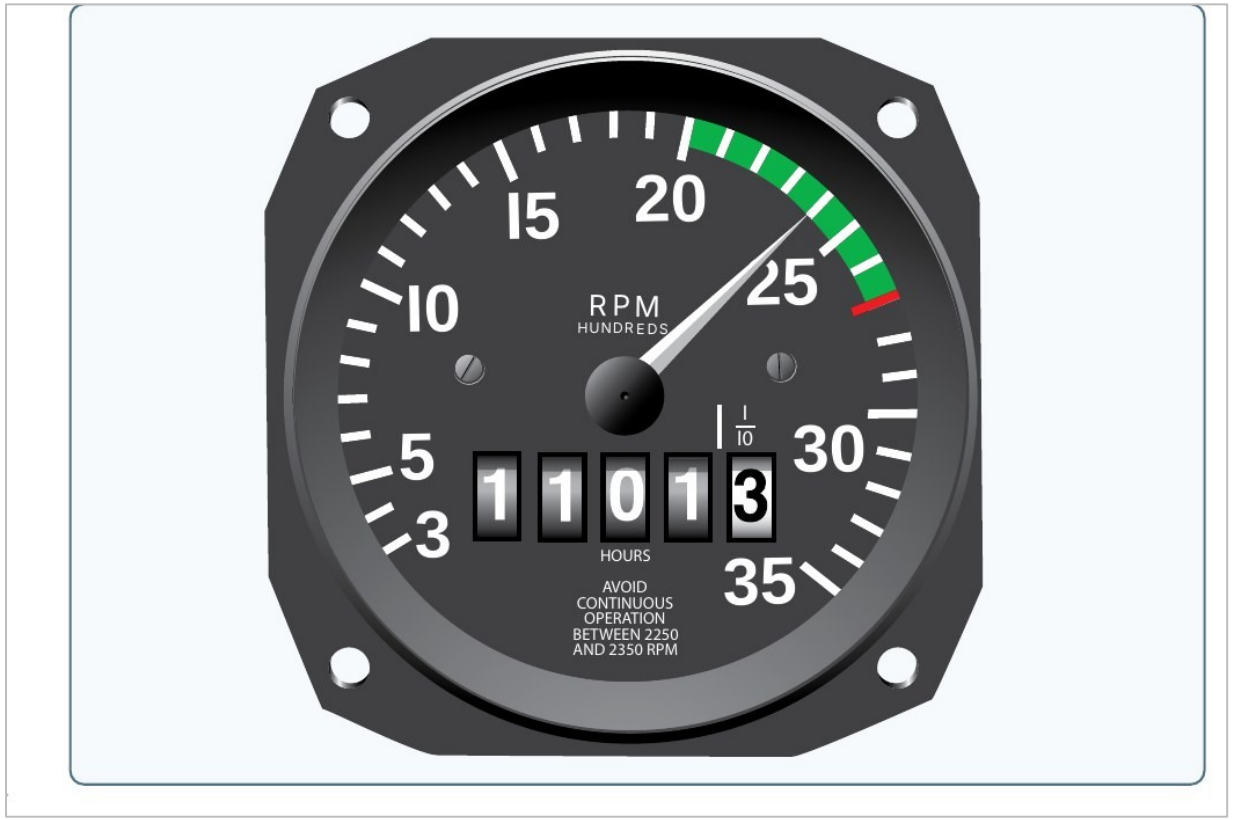
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Standard Conditions
Zero Wind

ALTITUDE	75% POWER		65% POWER		55% POWER	
	KTAS	NMPG	KTAS	NMPG	KTAS	NMPG
Sea Level	114	11.2	108	12.0	101	12.8
4000 feet	119	11.7	112	12.4	104	13.2
8000 feet	124	12.2	117	12.9	107	13.6

POH 4-33, PHAK 16-12

Power in a Fixed-Pitch Airplane

- Tachometer shows engine power
- Same RPM: less power up high
- Percent of max continuous power
- Above 75%: full rich



PHAK 7-5, POH 4-34, POH 4-35

Example: Read the Notes

- 8,000 ft, +20 C: 2,600 RPM
- 64% power, 117 KTAS
- Fuel flow: 8.9 GPH
- Note: above 75% full rich

CRUISE PERFORMANCE

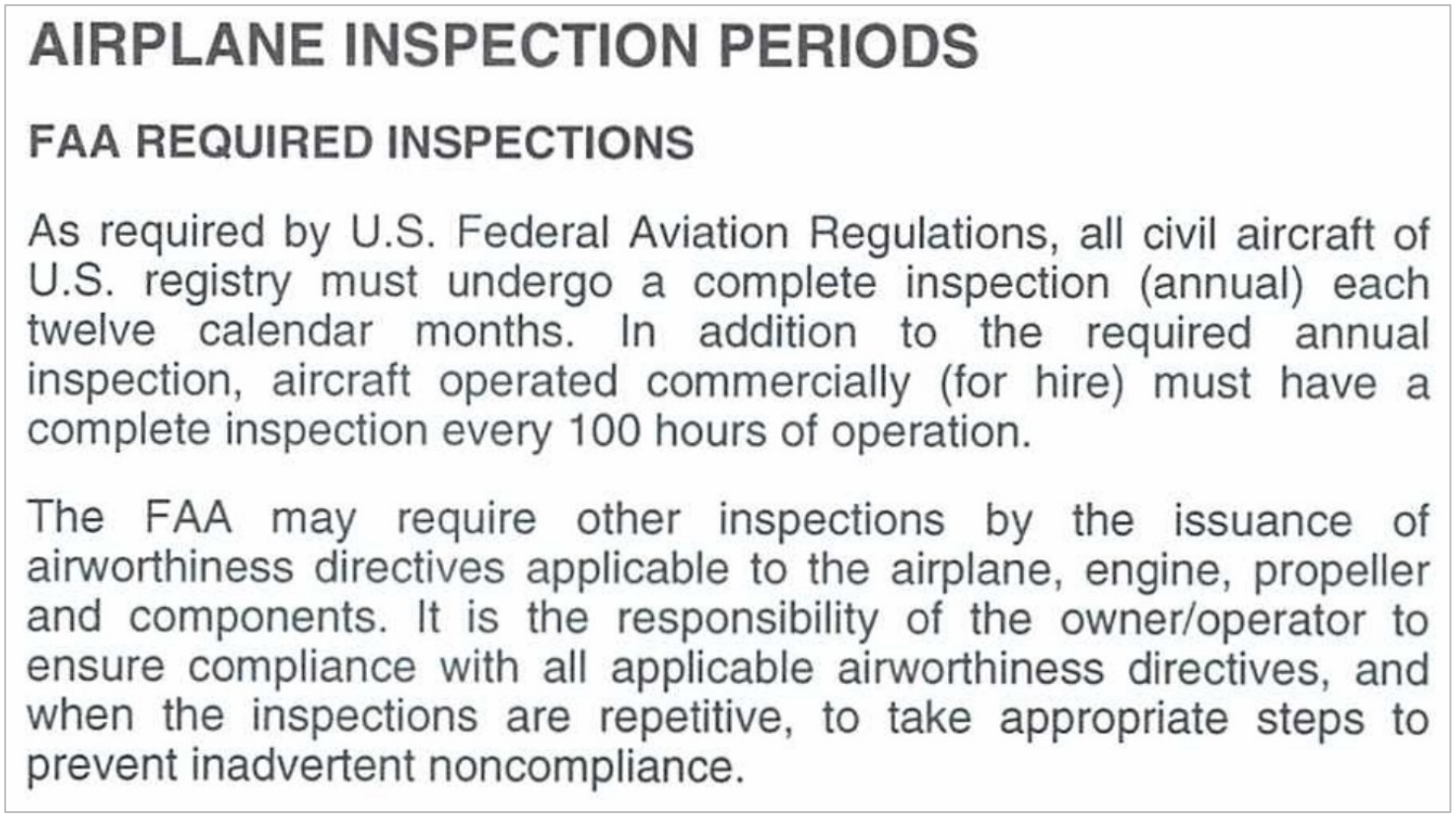
CONDITIONS:
2550 Pounds
Recommended Lean Mixture At All Altitudes (Refer to Section 4, Cruise)

PRESS ALT FT	RPM	20°C BELOW STANDARD TEMP			STANDARD TEMPERATURE			20°C ABOVE STANDARD TEMP		
		% MCP	KTAS	GPH	% MCP	KTAS	GPH	% MCP	KTAS	GPH
2000	2550	83	117	11.1	77	118	10.5	72	117	9.9
	2500	78	115	10.6	73	115	9.9	68	115	9.4
	2400	69	111	9.6	64	110	9.0	60	109	8.5
	2300	61	105	8.6	57	104	8.1	53	102	7.7
	2200	53	99	7.7	50	97	7.3	47	95	6.9
	2100	47	92	6.9	44	90	6.6	42	89	6.3
4000	2600	83	120	11.1	77	120	10.4	72	119	9.8
	2550	79	118	10.6	73	117	9.9	68	117	9.4
	2500	74	115	10.1	69	115	9.5	64	114	8.9
	2400	65	110	9.1	61	109	8.5	57	107	8.1
	2300	58	104	8.2	54	102	7.7	51	101	7.3
	2200	51	98	7.4	48	96	7.0	45	94	6.7
	2100	45	91	6.6	42	89	6.4	40	87	6.1
6000	2650	83	122	11.1	77	122	10.4	72	121	9.8
	2600	78	120	10.6	73	119	9.9	68	118	9.4
	2500	70	115	9.6	65	114	9.0	60	112	8.5
	2400	62	109	8.6	57	108	8.2	54	106	7.7
	2300	54	103	7.8	51	101	7.4	48	99	7.0
	2200	48	96	7.1	45	94	6.7	43	92	6.4

POH 5-5, POH Fig 5-8

Required Inspections

- Annual: 12 calendar months
- 100-hour: for hire, instruction
- May exceed 10 hours to shop
- Comply with all ADs



AIRPLANE INSPECTION PERIODS

FAA REQUIRED INSPECTIONS

As required by U.S. Federal Aviation Regulations, all civil aircraft of U.S. registry must undergo a complete inspection (annual) each twelve calendar months. In addition to the required annual inspection, aircraft operated commercially (for hire) must have a complete inspection every 100 hours of operation.

The FAA may require other inspections by the issuance of airworthiness directives applicable to the airplane, engine, propeller and components. It is the responsibility of the owner/operator to ensure compliance with all applicable airworthiness directives, and when the inspections are repetitive, to take appropriate steps to prevent inadvertent noncompliance.

14 CFR 91.409, 14 CFR 91.403, PHAK 9-8, POH 8-7

Other Required Checks

- Transponder: 24 calendar months
- ELT: 12 calendar months
- Altimeter, static: 24 (IFR)
- VOR check: 30 days (IFR)

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14 CFR 91.413, 14 CFR 91.207, 14 CFR 91.171, PHAK 9-9

Logbook Documentation

- Work done, date, signature
- Total time, AD status
- Kept by owner or operator
- Some records stay with plane

AIRPLANE FILE (Continued)

To be carried in the airplane at all times:

1. Current Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.
2. Garmin G1000 Cockpit Reference Guide (190-00384-00 or subsequent revision).
3. Weight and Balance, and associated papers (latest copy of the Repair and Alteration Form, FAA Form 337, if applicable).
4. Equipment List.

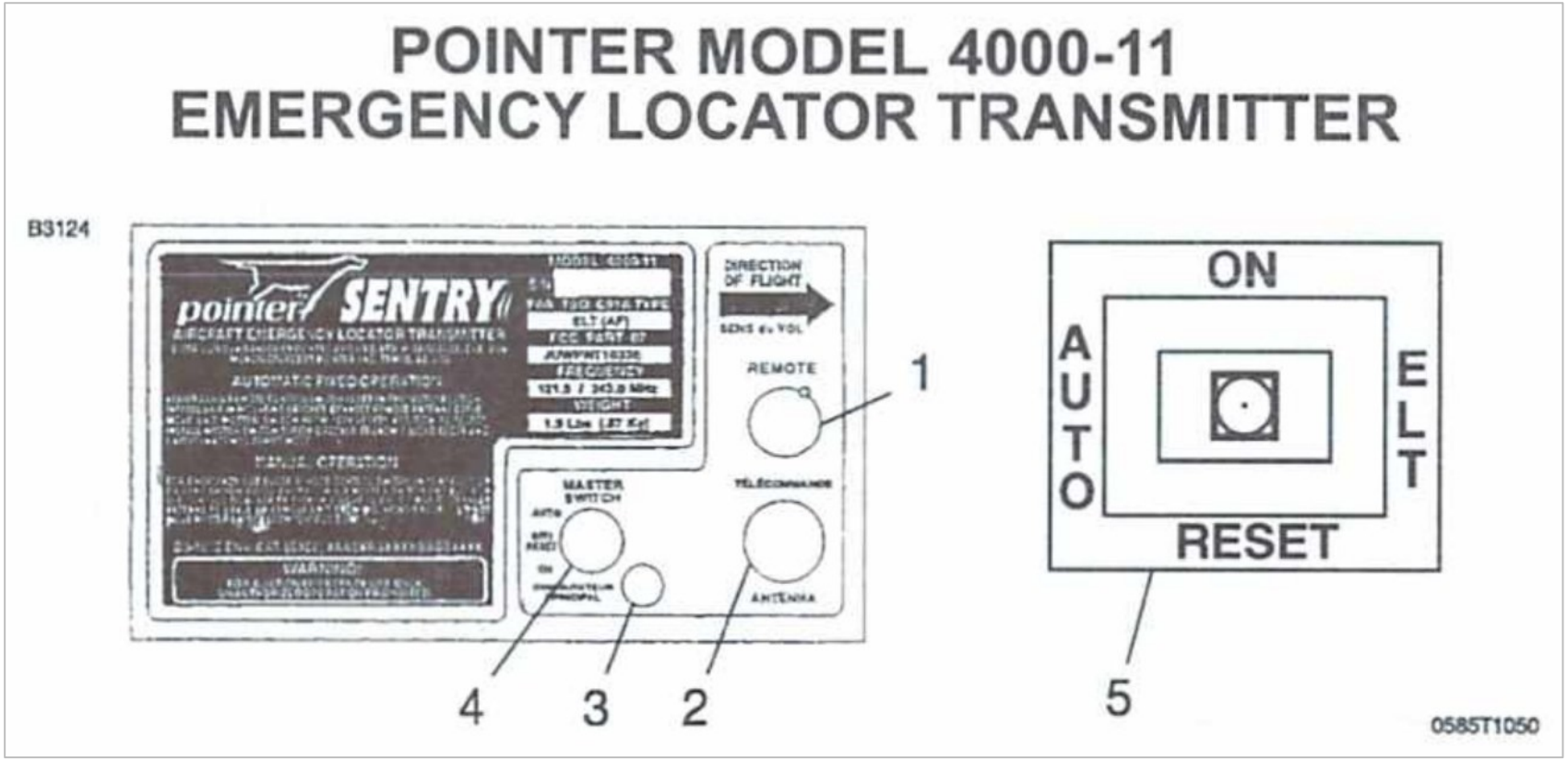
To be made available upon request:

1. Airplane Logbook.
2. Engine Logbook.

14 CFR 91.417, POH 8-7

ELT Operations

- 121.5, 243.0, or 406 MHz
- G switch auto activates
- Transmits at least 48 hours
- 406 MHz: register with NOAA



Using the C172S ELT

- Remote switch on panel
- Before forced landing: ON
- Rescue seen: RESET, AUTO
- After rescue: AUTO

CABIN FEATURES

EMERGENCY LOCATOR TRANSMITTER (ELT)

A remote switch/annunciator is installed on the top center location of the right front passenger instrument panel for control of the ELT from the flight crew station. The annunciator, which is in the center of the rocker switch, illuminates when the ELT transmitter is transmitting. The ELT emits an omni-directional signal on the international distress frequencies of 121.5 MHz and 243.0 MHz. General aviation and commercial aircraft, the FAA and CAP monitor 121.5 MHz, and 243.0 MHz is monitored by the military. For a basic overview of the ELT, refer to the Section 9, Supplement 1.

POH 7-71, POH S1-8, POH S1-9

ELT Limitations

- Satellites stopped 121.5 in 2009
- 121.5: needs ATC or aircraft
- Must survive the crash
- False alarms slow response

CABIN FEATURES

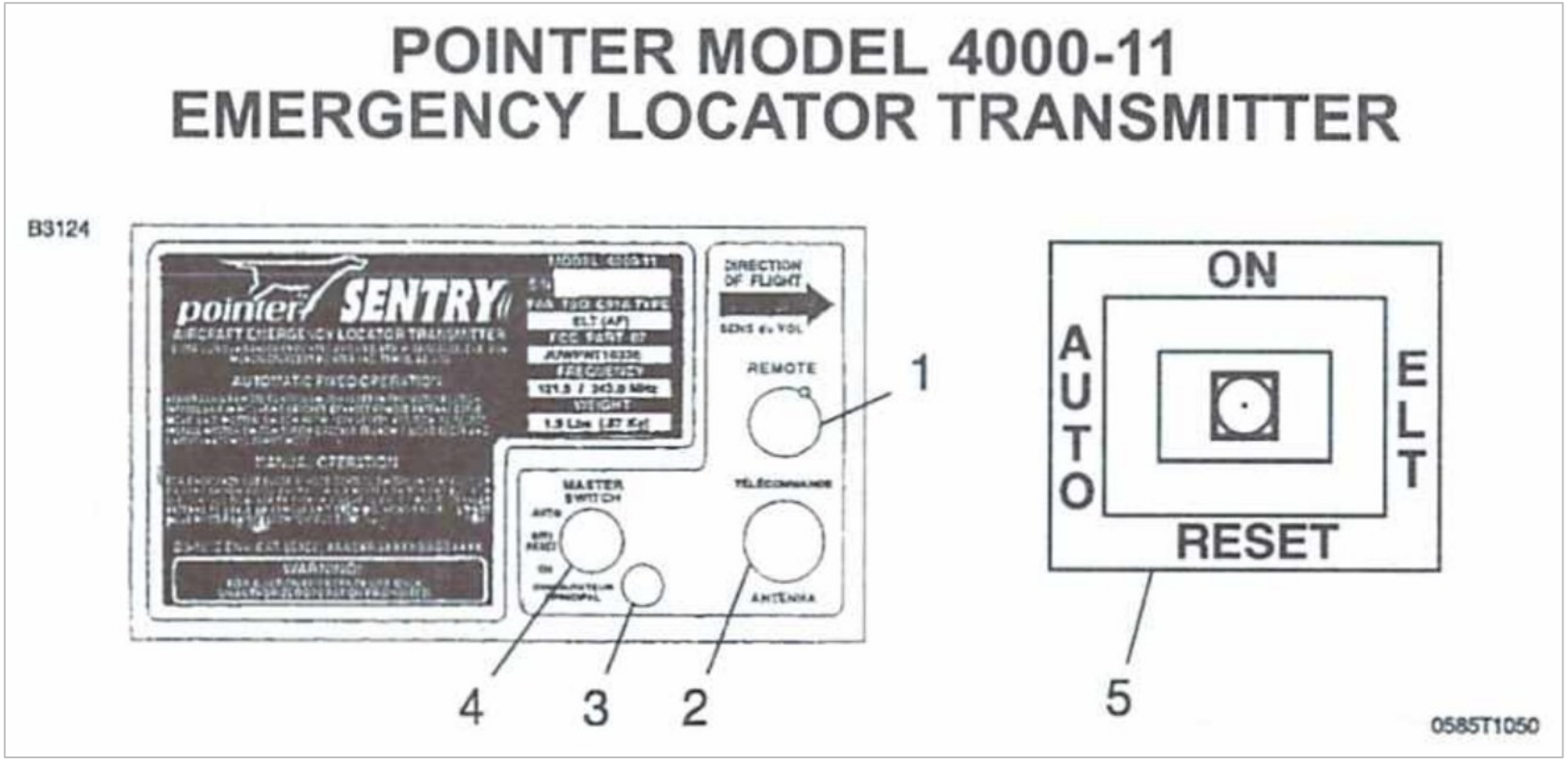
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AIM 6-2-4

ELT Testing and Rules

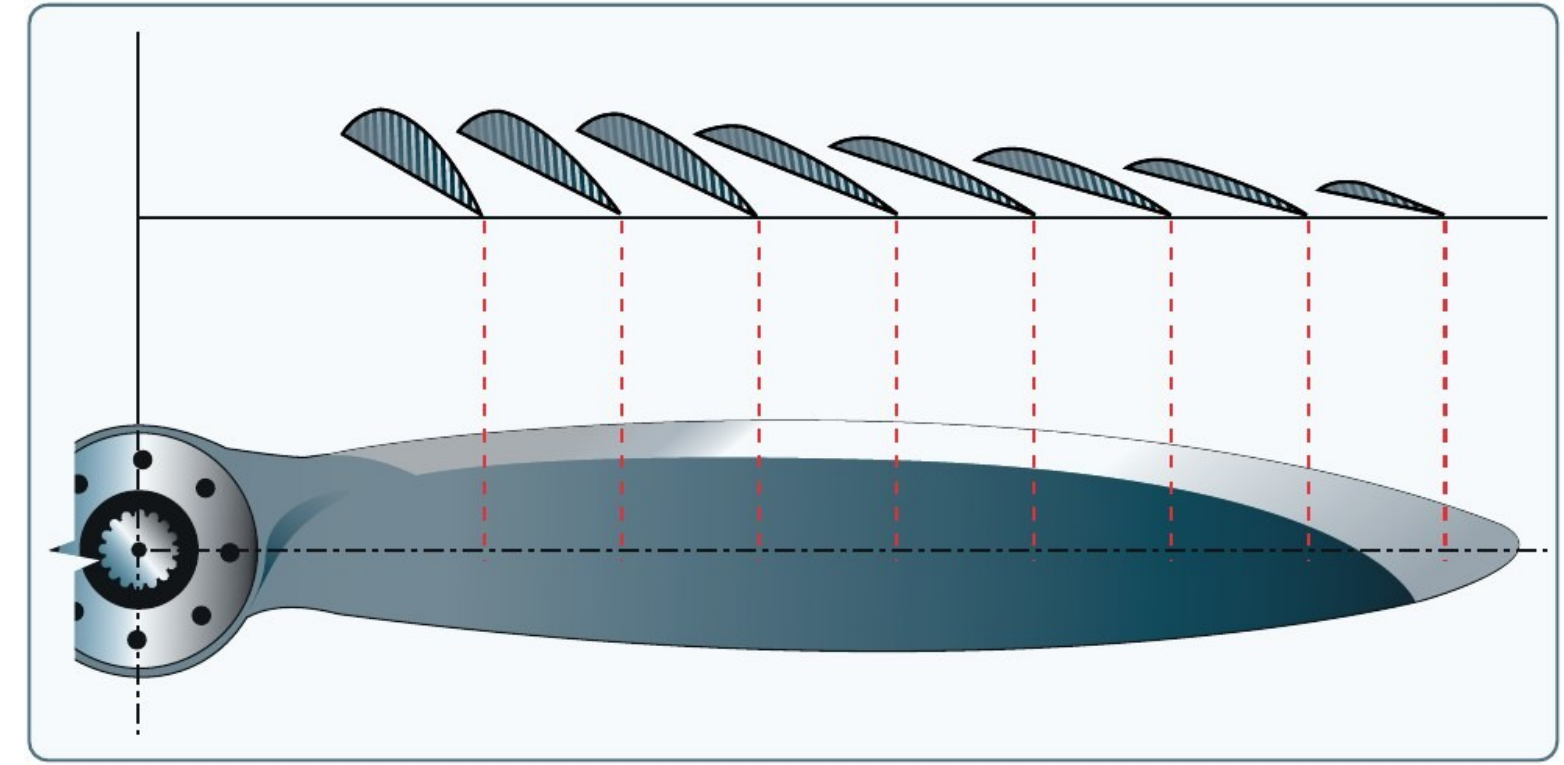
- Test first 5 minutes of hour
- No more than 3 sweeps
- Inspect every 12 calendar months
- Battery: 1 hour or 50% life



AIM 6-2-4, 14 CFR 91.207, POH S1-10

Study Guide Fixes

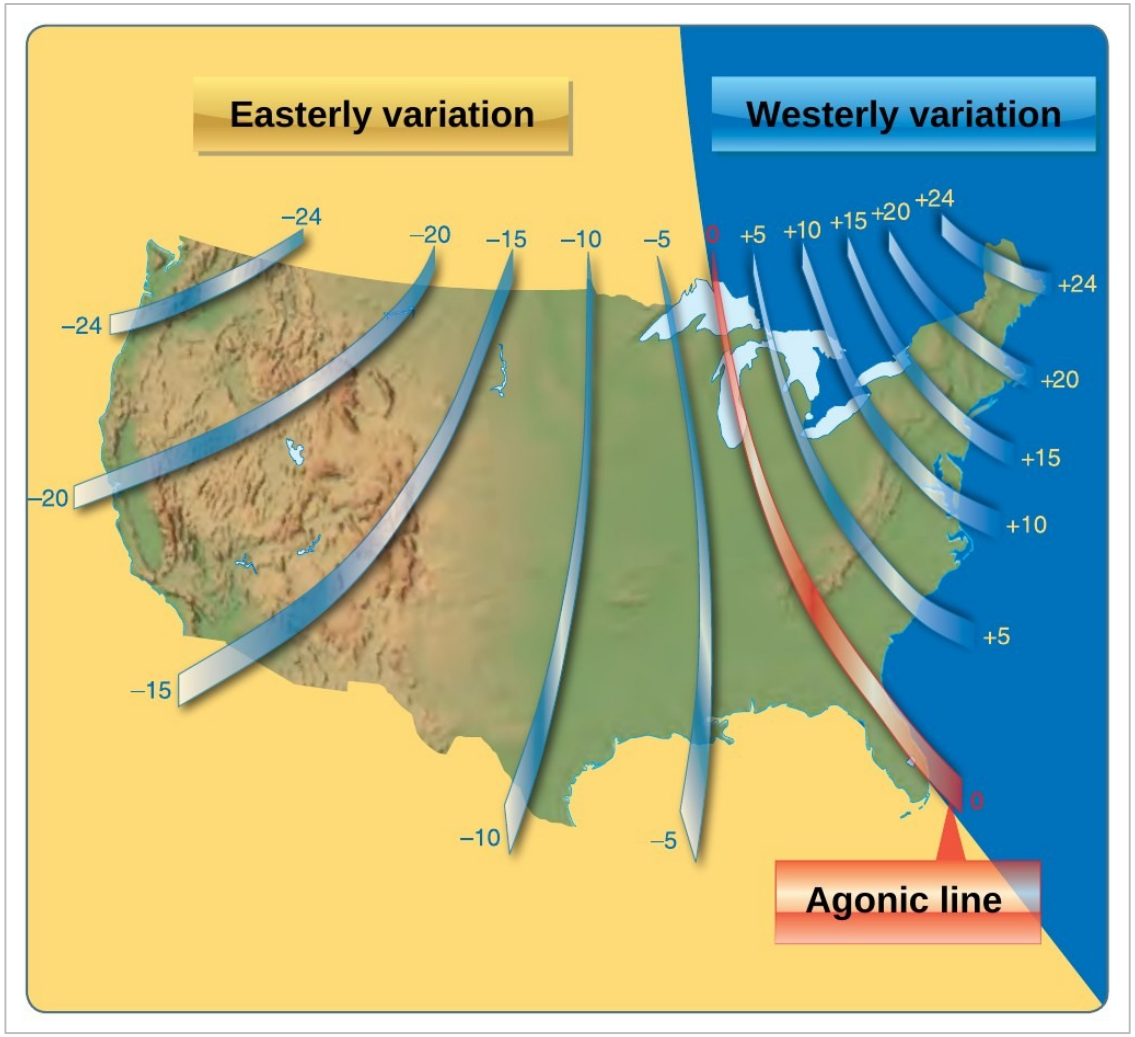
- Throttle: fuel-air flow, not fuel
- C172S: no propeller control
- Fix reverse sensing with OBS
- Use PHAK W&B wording



PHAK 7-5, POH 7-36, PHAK 16-26, PHAK 10-4, PHAK 5-29

Review Questions

- Name the perception factors
- Positive vs. neutral static?
- East variation: add or subtract?
- When may you test an ELT?



AIH 3-7, PHAK 5-14, PHAK 16-7, AIM 6-2-4

References

- AIH, PHAK, AIM
- 14 CFR Part 91
- C172S POH, TCDS 3A12
- FAA-S-ACS-25

CRUISE PERFORMANCE TABLE

CONDITIONS:
Standard Conditions
Zero Wind

ALTITUDE	75% POWER		65% POWER		55% POWER	
	KTAS	NMPG	KTAS	NMPG	KTAS	NMPG
Sea Level	114	11.2	108	12.0	101	12.8
4000 feet	119	11.7	112	12.4	104	13.2
8000 feet	124	12.2	117	12.9	107	13.6

See lesson plan for full list